

UGTS-KC 4.3

CHESS FIXED-POINT AND PROOF RUNTIME

Orthodox rule kernel, exact three-piece solution and full-solution proof architecture

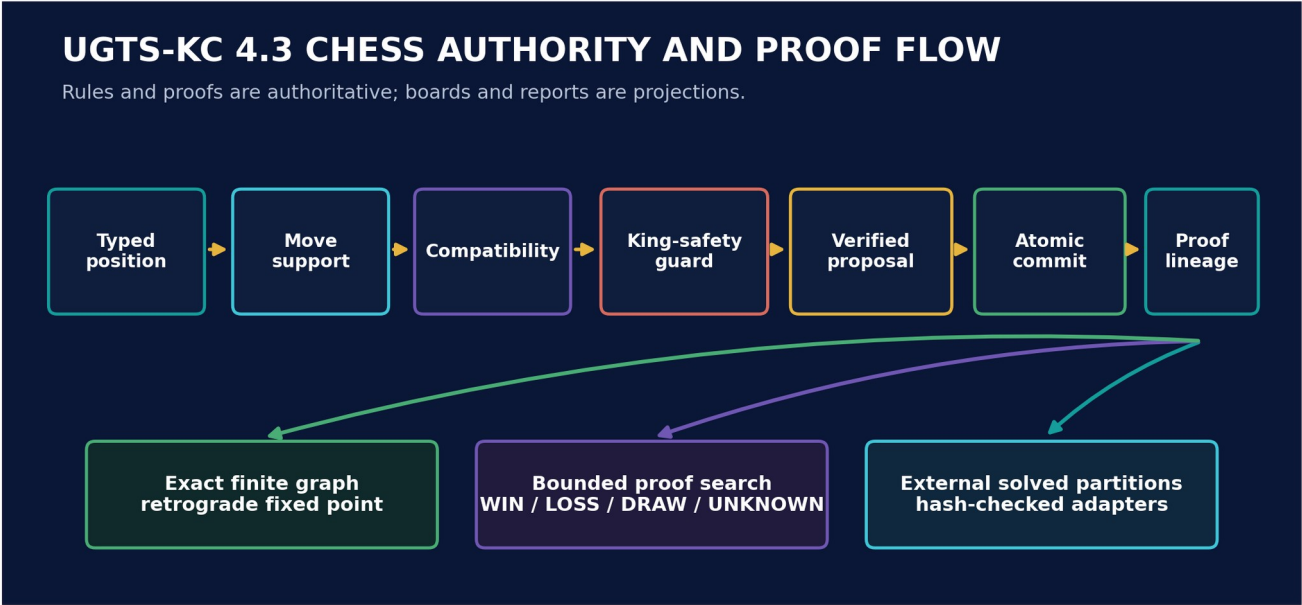


Figure 1. Chess is specialized into the canonical UGTS authority and proof flow.

RELEASE RESULT

The complete declared KQK, KRK, KBK, KNK and KPK three-piece profile is solved exactly. The orthodox 32-piece initial position is represented, tested and explicitly recorded as UNRESOLVED rather than assigned an unsupported result.

Prepared for	Tom Klootwijk
Version	4.3.0
Generated	29 August 2026
Release schema	ugts-kc-chess-4.3

1. Executive Decision and Solution Boundary

The word solve is treated as a proof obligation, not as a heuristic engine score.

DECISION - GO

Represent orthodox chess as a finite typed state-transition graph. Admit exact WDL/DTM labels only when all legal successor obligations are closed under a declared rule profile. Retain UNKNOWN/UNRESOLVED as first-class results when proof coverage is incomplete.

SOLUTION STATUS IS A TYPED RESULT

EXACT

Orthodox legal-move kernel; fixed-point solver; complete KQK, KRK, KBK, KNK and KPK profile; replay and hashes.

BOUNDED

Depth-limited proof search is sound inside its horizon. Common dead-material detection certifies selected exact cases only.

UNRESOLVED

Game-theoretic value of the 32-piece initial position and a complete all-material strategy certificate.

UNKNOWN is never silently rewritten as draw, equality or likely outcome.

Figure 2. Exact, bounded and unresolved results are separated in the release schema.

Measured release result

Item	Result	Meaning
Chess mechanisms	64 (M886-M949)	Eight typed domains; 52 executable reference records and 12 advanced contracts.
Exact tablebase	367,868 canonical states	Complete admitted KQK, KRK, KBK, KNK and KPK profile.
Graph coverage	3,217,270 directed edges	Every internal legal successor in the closed three-piece graph.
Binary payload	2,207,264 bytes	Compact exact WDL/DTM table plus section metadata.
Automated verification	31/31 tests PASS	Rule, perft, replay, fixed-point, tablebase, catalog and CLI checks.
Initial position	UNRESOLVED	Depth-three bounded proof visits 9,323 nodes and returns unknown.

A complete solution of the full initial position is not claimed. The release instead supplies a correct proof machine, an exact solved domain and a machine-readable route for extending proof coverage without silently promoting guesses.

2. Contents and Source Continuity

The chess profile is additive: it specializes retained definitions, order, event authority and replay.

Section	Purpose
1-3	Decision, scope, source basis and chess state identity.
4-7	Move authority, orthodox rules and legal-move validation.
8-12	Fixed-point proof calculus, tablebase construction and exact probes.
13-16	Replay, bounded proof search, operator catalog and scaling plan.
17-20	Kill criteria, reproduction, sources and final definition.

SOURCE CONTINUITY INTO THE CHESS PROFILE



Figure 3. UGTS-KC 4.3 specializes the retained authority chain rather than replacing it.

Source-derived commitments

- UGTS-KC 4.2 supplies typed operators, fixed points, proof obligations, canonical exchange and deterministic state authority [S1].
- KC Elizabeth 3.9 supplies deterministic game state, fixed-step discipline, canonical snapshots, hashes and replay [S2].
- KC Two Hands 3.0 supplies verified proposals, atomic commit, checkpoints and divergence detection [S3].
- UGTS-KC 3.6 supplies content-addressed definitions, dependency closure and explicit operation order [S4].
- FIDE Laws, published perft counts and Syzygy are external rules/validation/adaptor sources [S5-S7].

EVIDENCE BOUNDARY

Source continuity justifies the architecture. It does not by itself solve a chess state. Every game-theoretic label must still be discharged by legal graph coverage or a complete proof certificate.

3. Formal Chess State and Identity

A board diagram is a projection. Complete state includes rights, clocks and lineage.

q = (B, s, C, e, h, m, R, P, L)

B : 64-square occupancy map
C : castling rights
h : halfmove clock
R : repetition lineage
L : event and proof lineage

s : side to move
e : en-passant relation
m : fullmove number
P : rule / solver profile

Field	Typed meaning	Why it matters
B	Piece code or empty on each square	Defines geometry and occupancy but not the complete legal state.
s	White or Black	Changes which moves and attacks are authoritative.
C	Subset of KQkq	Two identical placements can differ in castling legality.
e	Square or none	Retained only when created by a double pawn move; repetition key keeps it only when a legal capture exists.
h / m	Non-negative clocks	Drive claimable/automatic draw status and exchange semantics.
R	Canonical repetition multiset/order	Coordinates alone cannot certify repeated-position conditions.
P / L	Versioned rule profile and lineage	Freeze semantics, proof scope, transitions and audit history.

Canonical exchange

Initial FEN:
rnbqkbnr/pppppppp/8/8/8/PPPPPPPP/RNBQKBNR w KQkq - 0 1

Move exchange: e2e4 | promotion: a7a8q

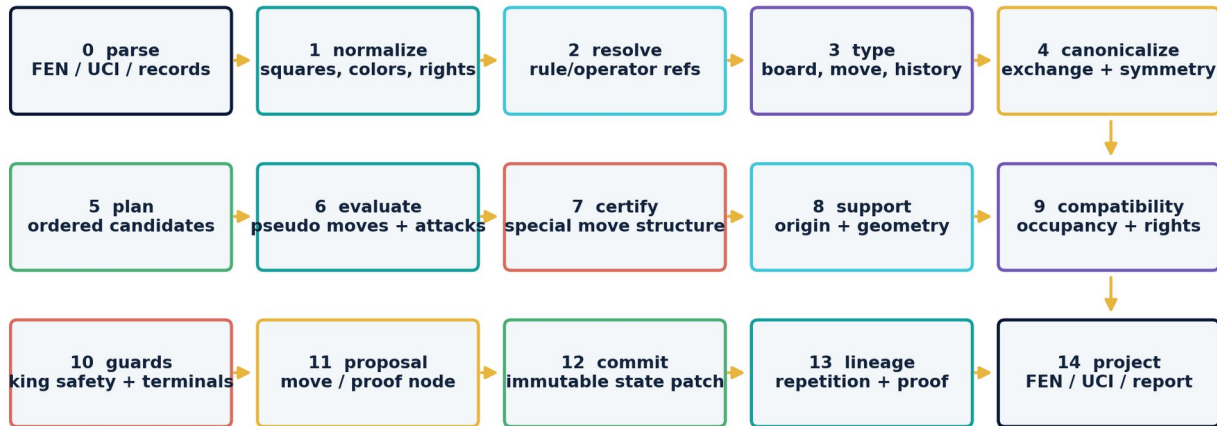
IDENTITY INVARIANT

The same board placement with a different side, castling right, legal en-passant capture, halfmove clock or repetition history may have a different successor relation or game-theoretic value.

4. Canonical Move Authority and Operation Order

The 4.2 phase schedule is specialized rather than hidden inside a move generator.

FIFTEEN CANONICAL PHASES SPECIALIZED FOR CHESS



Each phase is serialized and traceable; projection never mutates authoritative state.

Figure 4. Fifteen explicit phases separate parsing, construction, guards, commit, lineage and projection.

Authoritative move predicate

```

verified_move = origin_support_ok
    and piece_geometry_ok
    and occupancy_compatibility_ok
    and special_rights_ok
    and king_safety_guard_ok
    and pre_hash_matches_authority
    and numeric_error <= event_margin
  
```

Candidate generation may be parallel or heuristic. State mutation is not. A move becomes authoritative only after deterministic verification against the current immutable position and rule profile.

NON-COMMUTATIVE ORDER

Applying a move before the king-safety guard, changing castling rights after repetition hashing, or normalizing en-passant state without checking legal capture availability changes semantics. Order is part of the proof.

5. Orthodox Rule Kernel and Draw Semantics

The dependency-free kernel handles ordinary moves and the stateful exceptions that often break solvers.

Rule family	Implemented relation / guard	Boundary
Pawn	Single/double push, diagonal capture, en passant, four promotions	En-passant discovered-check is rejected by post-move king safety.
Knight	Eight bounded jumps	Destination must be empty or hold an opponent non-king piece.
Bishop/Rook/Queen	Ordered rays with first blocker	No move may jump through an occupied square.
King	Eight neighbors plus castling	Destination and castling transit squares must not be attacked.
Check/mate	Opponent attack on king + legal successor set	Checkmate is no legal move while in check; stalemate is no legal move while not in check.
Repetition	Canonical placement, side, castling, legal ep possibility	Threefold is claimable; fivefold is automatic under the selected FIDE profile.
Move clock	Pawn/capture resets, otherwise increment	Fifty moves is claimable; seventy-five moves is automatic; checkmate takes precedence.
Dead position	Selected exact material certificates	True is exact; false is not a proof that mate is possible.

Terminal precedence

1. Generate all legal moves.
2. If none: checkmate if in check, otherwise stalemate.
3. Evaluate exact dead-position certificates.
4. Evaluate fivefold / seventy-five-move automatic draws.
5. Expose threefold / fifty-move claim actions as claimable status.

RULE PROFILE BOUNDARY

The generic full game graph must model claimable draws as explicit player actions. The exact bundled three-piece tablebase uses a zero-clock, no-castling profile so its graph remains finite and closed under the declared semantics.

6. Exact Legal-Move Validation

Perft counts validate the successor relation without relying on an evaluation function.

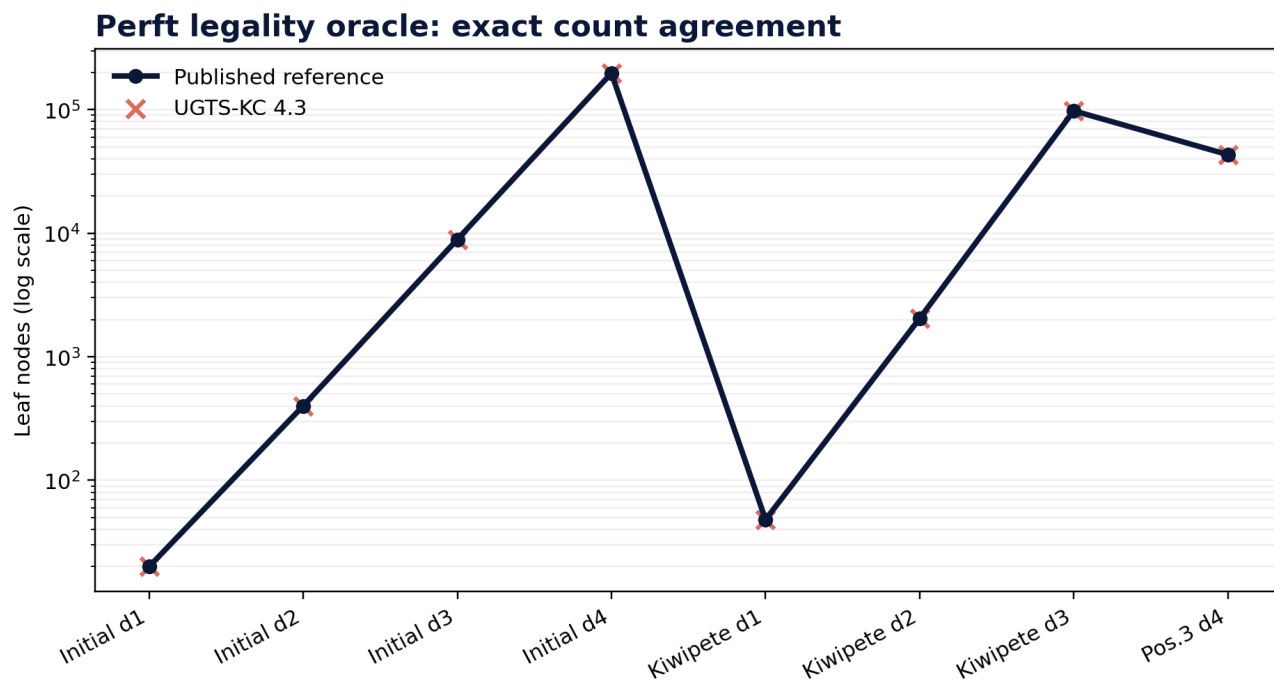


Figure 5. UGTS-KC 4.3 matches published leaf-node counts on the tested depths.

Position	Depths checked	Expected / actual highest depth	Coverage stress
Initial	1-4	197,281 / 197,281	Ordinary development, checks and early captures.
Kiwipete	1-3	97,862 / 97,862	Castling, pins, checks and dense tactical branches.
Position 3	1-4	43,238 / 43,238	En passant, checks, rook/pawn geometry and unusual king exposure.
Position 4	1-3	9,467 / 9,467	Castling rights, promotions and tactical legality.

Validation interpretation

Perft agreement is a strong legality regression test: it checks the exact number of legal move paths. It does not prove strategic correctness by itself, but a fixed-point proof built on an incorrect successor relation would be invalid, so this gate is mandatory.

CURRENT CAPTURED RESULT

31 unit tests pass. Compile checks, JSON parsing, catalog continuity, tablebase hashes and three independent perft cases also pass in the release validator.

7. Exact Fixed-Point Solution Calculus

Chess becomes solvable on any finite, complete and closed state graph.

EXACT WDL/DTM AS A LEAST FIXED POINT

Monotone propagation continues until no new W/L nodes are discovered.

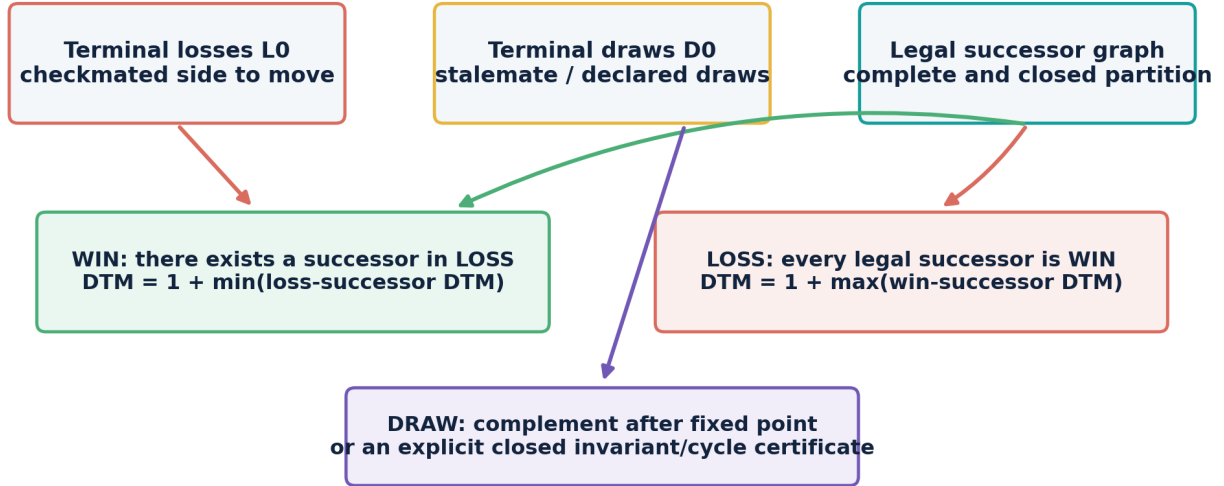


Figure 6. The least fixed point propagates win and loss from terminal losses; the remaining complement is draw.

```

W_(n+1) = W_n union { p | exists q in Succ(p): q in L_n }
L_(n+1) = L_n union { p | Succ(p) nonempty and every q in Succ(p) lies in W_(n+1) }

DTM(win p) = 1 + min DTM(loss successor)
DTM(loss p) = 1 + max DTM(win successor)
  
```

Values are always relative to the player to move. A strong-side win is derived afterward from side-to-move value, color normalization and material ownership. This avoids silently confusing a losing side-to-move node with a losing material class.

CLOSURE REQUIREMENT

The draw complement is exact only when the partition is complete and closed, all terminal rules are explicit, and every legal successor has been represented. Otherwise the correct result is UNKNOWN.

8. Proof Obligations and Certificate Semantics

A proof is not a principal variation alone; obligations differ by value.

PROOF OBLIGATIONS ARE DIFFERENT FOR WIN, LOSS AND DRAW

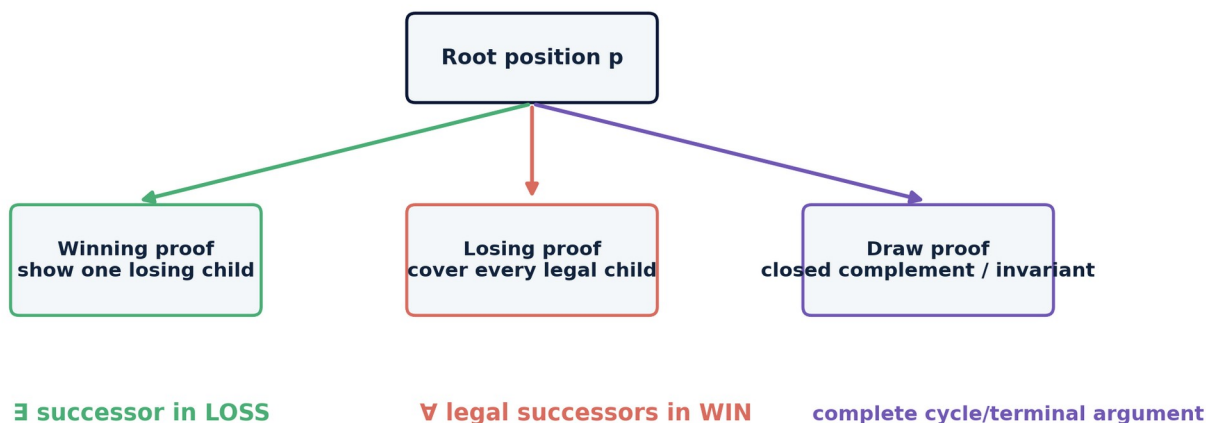


Figure 7. Winning, losing and drawn nodes require different quantified evidence.

Claim	Required certificate	Invalid shortcut
WIN	At least one verified legal move to a LOSS node, recursively certified.	A positive heuristic score or one attractive line.
LOSS	Every verified legal move leads to a WIN node, with complete successor coverage.	Showing only the opponent best reply.
DRAW	Solved complement, closed invariant, or complete cycle/terminal argument under the rule profile.	No mate found at a finite search horizon.
UNKNOWN	Proof budget or graph coverage ended before an obligation closed.	Silently coercing the result to draw.

Certificate identity

```
certificate_hash = SHA256(canonical JSON of
  root hash + rule profile + solver profile + obligations + child hashes + status)
```

A hash detects change/divergence; full proof content remains authoritative.

REPLAY AND PROOF DAG SHARING

Transpositions share canonical nodes. Event replay checks pre/post hashes; proof replay checks child obligations and content hashes. Neither digest replaces the underlying state, legal-move set or lineage.

9. Exact Three-Piece Tablebase Construction

The release solves a complete symmetry-reduced graph, not a sampled set of positions.

- 1. Enumerate legal KQK, KRK, KBK, KNK and KPK states with the extra piece normalized to White.
- 2. Reject adjacent kings, missing kings, impossible occupancy and illegal predecessor/check states.
- 3. Normalize non-pawn positions under the eight D4 board symmetries; normalize pawn positions by file reflection only.
- 4. Generate every legal successor, including pawn promotion into queen, rook, bishop or knight sections.
- 5. Build exact predecessor lists and terminal checkmate seeds.
- 6. Run the monotone W/L fixed point; classify the unmarked closed complement as draw.
- 7. Propagate DTM with min on wins and max on losses.
- 8. Write sorted canonical keys and compact values; probe through the same canonicalization contract.

Profile field	Value
Profile ID	orthodox-three-piece-zero-clock-no-castling-v1
Material classes	KQK, KRK, KBK, KNK, KPK with all promotions
Color normalization	Extra piece belongs to White; color-swapped positions are isomorphic
Symmetry	D4 for Q/R/B/N; file reflection for P
Canonical states	367,868
Internal directed edges	3,217,270
Terminal checkmates	73
Build evidence	16.766 s in the captured run; deterministic sorted output

SCOPE BOUNDARY

Castling rights are absent, the halfmove clock begins at zero, and the table is not promoted as a solution of the 32-piece initial position. The profile is explicit in every probe result.

10. Exact Three-Piece Results

Every value below is generated from the closed graph and stored in the bundled binary tablebase.

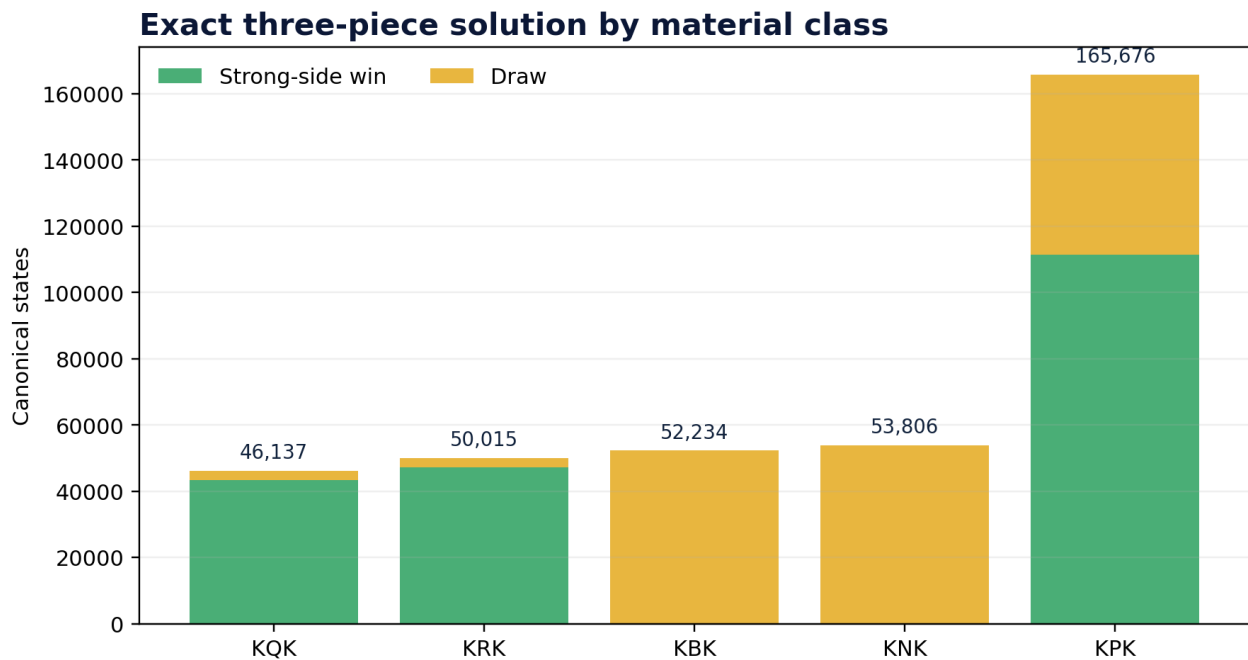


Figure 8. Exact strong-side wins and draws by material class.

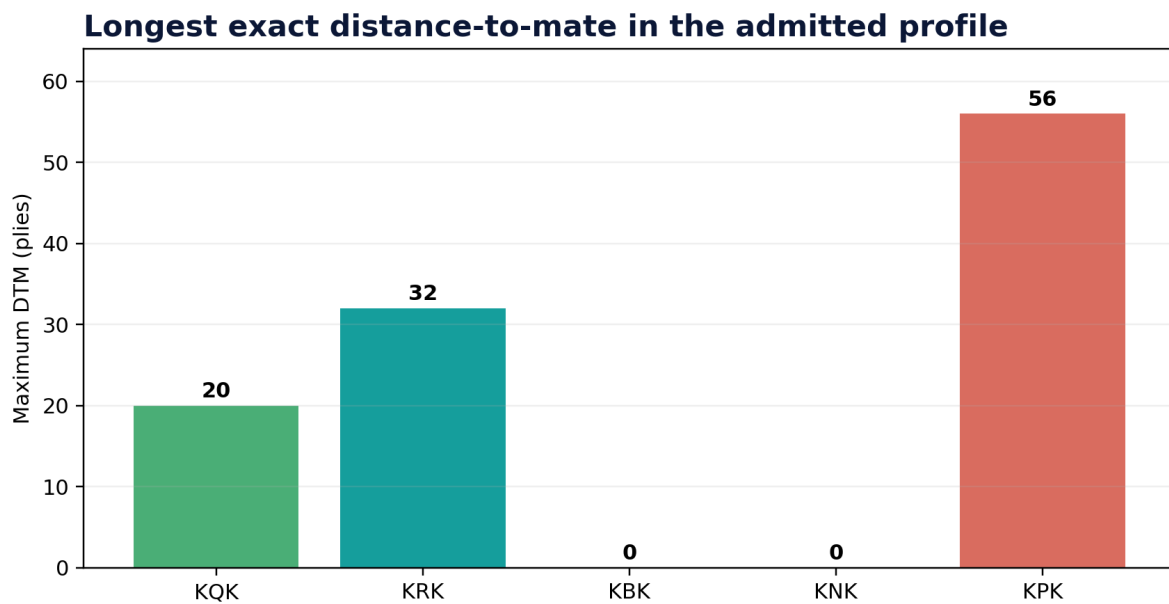


Figure 9. Maximum exact distance to mate in plies.

Class	States	Strong-side win	Draw	Max DTM
KQK	46,137	43,241	2,896	20
KRK	50,015	47,219	2,796	32
KBK	52,234	0	52,234	0
KNK	53,806	0	53,806	0
KPK	165,676	111,282	54,394	56

11. Worked Exact Probes

These positions are witnesses for the longest DTM values in the shipped profile.

Exact longest-line witnesses in the bundled tablebase

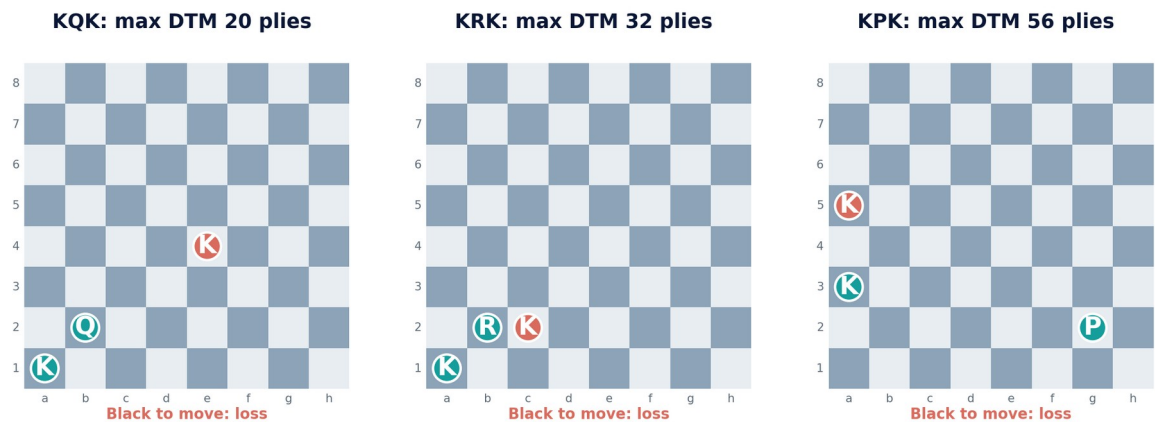


Figure 10. Exact KQK, KRK and KPK longest-line examples after canonicalization.

FEN	Side-to-move value	Strong-side value	DTM / best move
8/8/8/8/4k3/8/1Q6/K7 b - - 0 1	LOSS	White WIN	20 plies; e4f5 delays mate
8/8/8/8/8/8/1Rk5/K7 b - - 0 1	LOSS	White WIN	32 plies; exact longest KRK class
8/8/8/k7/8/K7/6P1/8 b - - 0 1	LOSS	White WIN	56 plies; exact longest KPK class

Probe semantics

- `side_to_move_value` is the local minimax value at the queried node.
- `strong_side_value` translates the local result back to the side owning the extra piece.
- `dtm_plies` is exact under the table profile; best moves maximize survival in LOSS and minimize mate length in WIN.
- A successful probe returns the canonical key, material section, color-swap flag and profile boundary.

DRAWN MATERIAL CLASSES

Every admitted legal KBK and KNK state is drawn. KQK and KRK still contain drawn states where the extra piece is capturable, lost, or the position terminates without mate.

12. Move Proposals, Atomic Commit and Replay

Search, tablebases and humans may propose; one deterministic boundary changes state.

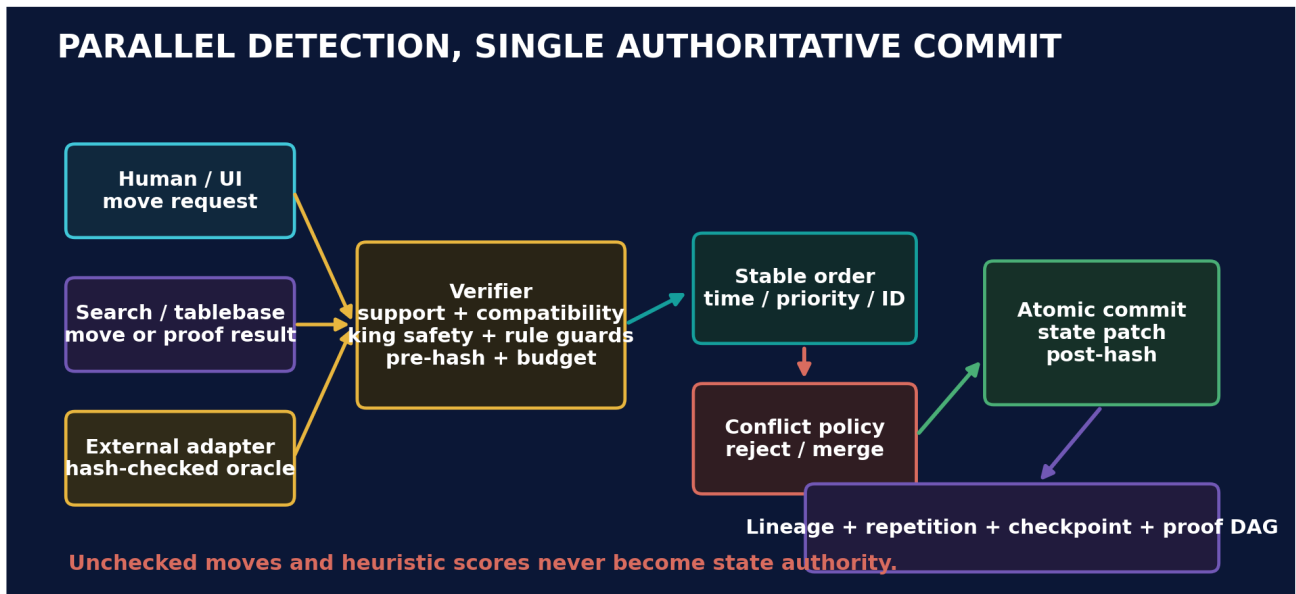


Figure 11. Parallel detection feeds a single verified, ordered and hash-checked commit path.

Runtime event record

```

RuntimeEvent = {
  sequence, proposal_id, uci, source,
  pre_hash, post_hash, repetition_key,
  halfmove_clock, fullmove_number,
  terminal_kind, winner, event_hash
}
  
```

The demonstration commits e2e4, e7e5, g1f3, b8c6 and f1b5. Replay starts from the initial position, recomputes every legal move and halts at the first pre-hash, post-hash or event-hash mismatch.

AUTHORITY INVARIANT

Unchecked GPU writes, stale tablebase probes, UI animations and heuristic lines never mutate the chess state. A proposal whose pre-hash does not match the current authority is rejected.

13. Bounded Proof Search and Initial-Position Status

Finite-horizon search remains useful when its epistemic boundary is machine-readable.

Example	Depth	Nodes	Result	Certificate
Mate in one: 7k/8/6K1/8/8/5Q2/8 w - - 0 1	1	30	WIN in 1 ply	Best move f2f8; terminal proof closed.
Orthodox initial position	3	9,323	UNKNOWN	No full-game value inferred.

```
if terminal: return exact terminal value
if depth == 0: return UNKNOWN
if any child is LOSS: return WIN
if all children are WIN: return LOSS
if all children are WIN or DRAW and at least one DRAW: return DRAW
otherwise: return UNKNOWN
```

INITIAL-POSITION RELEASE STATUS

UNRESOLVED under full strong-solution scope. The position is fully represented, legal moves are validated, and a bounded proof certificate is emitted. This is an explicit result, not a refusal or a hidden heuristic.

Why depth-limited no-mate is not draw

A draw claim quantifies over all possible future play or establishes a closed repetition/terminal invariant. Reaching a search horizon only proves that the selected proof budget ended. The runtime therefore keeps UNKNOWN separate from DRAW in APIs, JSON and tests.

14. Operator Catalog M886-M949

Chess receives a contiguous 64-mechanism namespace without renumbering the retained 4.2 catalog.

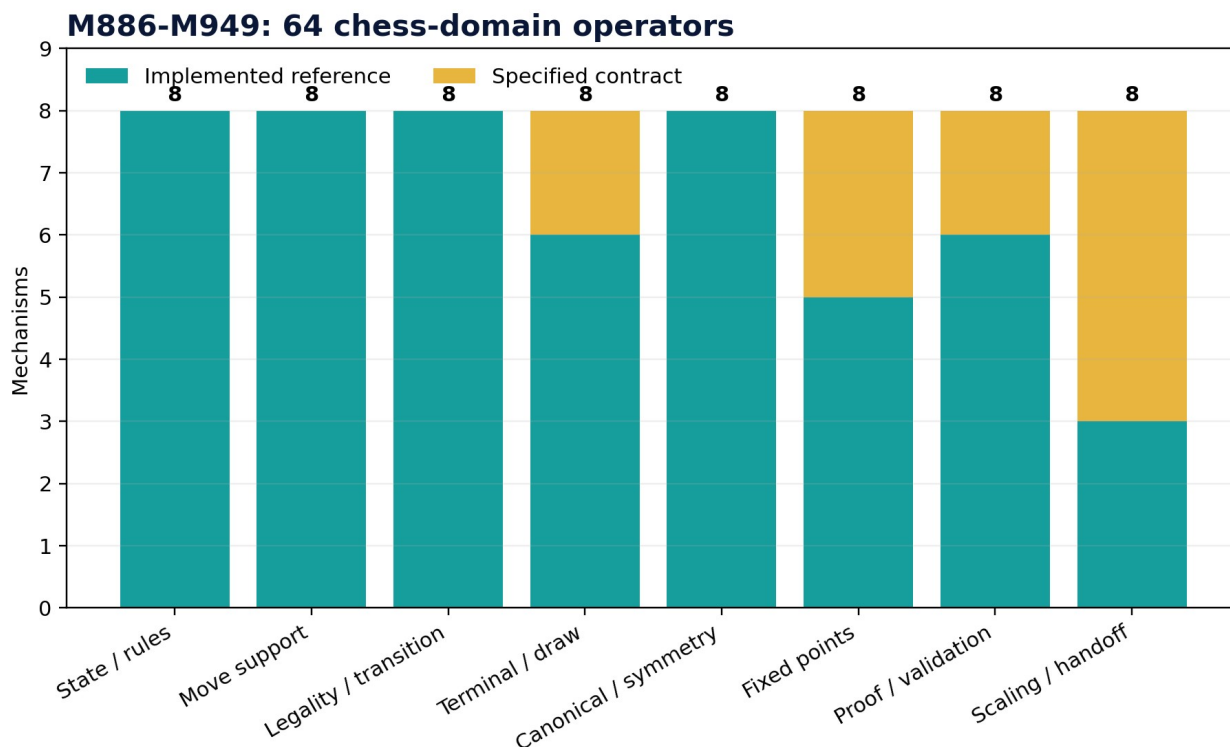


Figure 12. Eight domains contain eight mechanisms each; implemented and contract-only status stays visible.

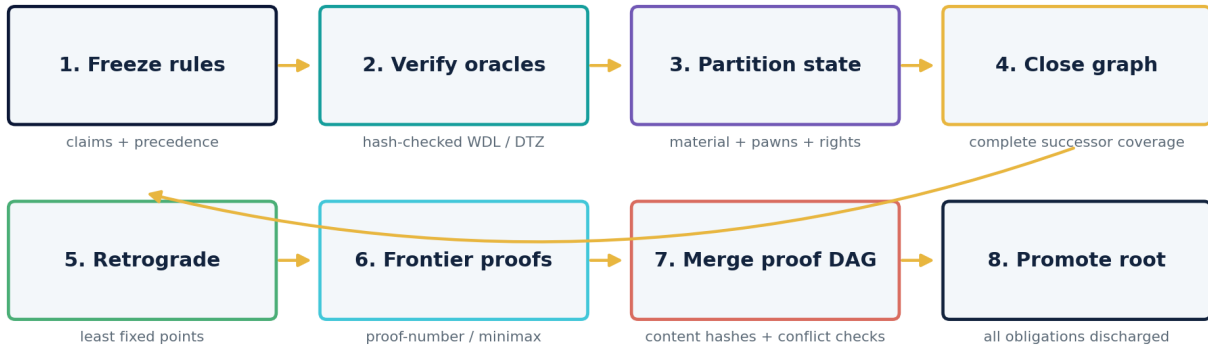
Range	Domain	Representative mechanisms
M886-M893	Chess state and rule schema	Rule profile, squares, pieces, occupancy, side, castling, ep, clocks and repetition.
M894-M901	Move geometry and support	Pawn, knight, sliders, king, castling path, attacks and first blocker.
M902-M909	Legality and transitions	Pseudo moves, support, compatibility, king safety, proposals and atomic transition.
M910-M917	Terminal and draw calculus	Check, mate, stalemate, dead-position contract, clocks, repetition and precedence.
M918-M925	Canonicalization and symmetry	FEN/UCI, repetition keys, hashes, D4, color swap and transposition DAG.
M926-M933	Game solving and fixed points	Finite graph, minimax, retrograde, WDL, DTM, bounded proof and frontier schedule.
M934-M941	Proof, validation and status	Proof nodes, strategy certificates, perft, conformance, replay and unresolved status.
M942-M949	Scaling and UGTS handoff	Packed profiles, partitioning, joins, compression, Syzygy/GPU contracts and kill criteria.

Catalog meaning: Mechanism IDs provide traceability, typed contracts and validation status. They are not by themselves claims of novelty, patentability or a solved initial chess position.

15. Scaling Toward a Full Strong Solution

The exact three-piece release is a base partition in a larger proof DAG.

PROMOTION PATH TO A FULL STRONG SOLUTION



Heuristic evaluation may order work; it cannot certify the root.

Figure 13. Promotion of the initial position occurs only after all quantified move obligations are discharged.

External tablebase adapter

The M946 contract accepts external Syzygy WDL/DTZ partitions only after profile, material, file checksum, probe result and 50-move semantics are verified. The public Syzygy generator supports endgame tablebases through seven pieces, but those datasets are not redistributed in this compact release [S7].

Full-solution work partition

- Partition by material, pawn structure, castling rights, en-passant state and symmetry orbit.
- Reject any slice with incomplete predecessor or successor coverage.
- Use exact retrograde on closed slices and proof-number/minimax frontier work on open higher-material regions.
- Share transpositions through canonical content hashes and merge only under deterministic conflict policy.
- Use heuristic engines solely to order work; never as proof authority.

PROMOTION GATE

The initial position receives WIN, LOSS or DRAW only when its root obligation and every required descendant obligation reference complete verified certificates under the frozen rule profile.

16. Storage, Error Contracts and Kill Criteria

Compression is admitted only when semantics, decode cost and proof reconstructibility remain explicit.

Record	Current release	Boundary
Three-piece table	2,207,264-byte binary; SHA-256 checked	Exact only for the declared zero-clock/no-castling profile.
Position hash	Canonical JSON SHA-256	Divergence/integrity signal, not a substitute for full state or legal proof.
Repetition key	Placement + side + rights + legally relevant ep	Clocks excluded by repetition law; lineage counts remain external.
Symmetry key	D4 or file reflection canonical representative	Only transformations preserving piece movement and profile semantics are admitted.
Proof DAG	Content-addressed obligations and child hashes	A compressed certificate must still reconstruct every quantified obligation.
External oracle	Profile + checksum + probe result	Rejected when rule semantics, files or successor consistency do not match.

Solver kill criteria

- A legal move, predecessor or terminal action is missing from a claimed closed partition.
- Castling, en-passant, repetition or draw semantics are implicit or disagree across joined partitions.
- Canonicalization merges non-isomorphic states or a content hash fails verification.
- Compression or packed precision changes move legality, event order, WDL or DTM.
- A heuristic score, principal variation or finite search horizon is promoted as a strong proof.
- Graph/frontier density, merge conflicts or proof storage exceed the declared budget without an explicit UNKNOWN result.
- A conventional verified solver is simpler, faster or more auditable at equal proof coverage.

EVIDENCE RULE

Failure of a promotion gate does not prove draw or impossibility. It returns a reason-coded incomplete/unknown status and preserves the last verified boundary.

17. Package Layout, Reproduction and Validation

The source, table, examples, tests and evidence are shipped together.

Path	Contents
src/ugts_chess43/	Immutable position/move records, rules, canonical identities, proposal/commit/replay, search and three-piece tablebase.
data/	Exact binary tablebase and machine-readable build summary.
spec/	Formal definition, M886-M949 CSV/JSON catalog, rule profiles, schema and source register.
examples/	Initial UNKNOWN certificate, mate-in-one proof, event replay and exact tablebase probe.
tests/	31 tests spanning move rules, perft, special moves, terminal states, replay, fixed points, tablebase, catalog and CLI.
validation/	Captured test output, perft checks, compile/JSON gates and hashes.
report/	This editable DOCX and derived PDF.
checksums/	SHA-256 manifest and release summary.

Reproduction commands

```
cd UGTS_KC_4_3_CHESS
PYTHONPATH=src python -m unittest discover -s tests -v
PYTHONPATH=src python examples/demo.py
PYTHONPATH=src python -m ugts_chess43 build-tb3 data/tb3_orthodox_zero_clock.bin --summary
data/tb3_summary.json
python tools/generate_spec.py
python tools/validate_release.py
```

Gate	Captured result
Python compile	PASS
Automated tests	31/31 PASS
Perft reference checks	PASS
Catalog continuity	M886-M949, 64 unique hashes
Tablebase integrity	2,207,264 bytes; SHA-256 16a9c6f7...c035
JSON/schema records	Parse and validation gates PASS

Reproducibility boundary: Regeneration is deterministic under the same specified Python semantics and sorted canonical key order. Cross-implementation identity still requires the same rule profile, canonicalization and binary format.

18. Source Register and Normative References

External standards are cited for rule semantics and validation; UGTS documents supply the architectural basis.

ID	Source	Release use
S1	UGTS-KC 4.2 General Operator and Order Addendum	Typed operators, fixed points, proof obligations, deterministic order and canonical exchange.
S2	UGTS-KC 3.9 KC Elizabeth Vector Game Runtime	Deterministic game state, snapshots, hashes, replay and headless validation.
S3	UGTS-KC Two Hands 3.0	Proposal verification, atomic commit, checkpoints and divergence detection.
S4	UGTS-KC 3.6 Literal Referential Substrate	Content-addressed definitions, dependencies and explicit operation order.
S5	FIDE Laws of Chess effective 1 January 2023	Orthodox move, terminal, repetition and move-count rule profile.
S6	Chess Programming Wiki - PerfT Results	Published legal move-path reference counts used in regression tests.
S7	syzygy1/tb - Chess endgame tablebase generator	External WDL/DTZ adapter target and seven-piece boundary.

Reference locations

[S5] <https://handbook.fide.com/chapter/e012023>

[S6] https://www.chessprogramming.org/Perft_Results

[S7] <https://github.com/syzygy1/tb>

Attribution and evidence notice

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External URLs and public chess reference data are not silently relicensed by this package. The included implementation and generated table are original engineering artifacts built for this release; see NOTICE.md for the package boundary.

SCIENTIFIC BOUNDARY

The release demonstrates executable internal consistency, exact results inside a declared finite profile and a falsifiable path toward larger proofs. It does not establish the game-theoretic value of the initial position.

19. Final Formal Definition and Release Decision

The solved object is a typed proof state, not a rendered board or an engine evaluation.

FORMAL DEFINITION

UGTS-KC 4.3 is a content-addressed orthodox-chess profile in which a complete position state defines a deterministic legal successor relation; every move passes through support, compatibility and king-safety guards; verified proposals commit immutable state patches with repetition and replay lineage; finite closed partitions are solved by a least W/L fixed point with exact DTM; bounded searches return UNKNOWN when obligations remain open; and only complete hash-checked proof certificates may promote a root to WIN, LOSS or DRAW.

```
UGTS-CHESS = (Q, M, A, G, T, R, V, C, L)

Q : typed position states
M : deterministic legal move relation
A : attack and king-safety relations
G : terminal and draw guards
T : atomic transitions
R : repetition and rule-profile state
V : WDL / DTM fixed-point values
C : proof certificates and validation evidence
L : event, replay and proof lineage
```

Release decision

- GO: the orthodox rule kernel, event authority and fixed-point proof runtime are coherent and executable.
- SOLVED: the complete declared KQK, KRK, KBK, KNK and KPK three-piece domain.
- BOUNDED: finite-horizon proof search and selected dead-material certificates.
- UNRESOLVED: the game-theoretic value of the orthodox 32-piece initial position.
- NEXT PROMOTION GATE: verified external endgame partitions and a content-addressed all-material proof DAG.

FINAL STATEMENT

Chess has been converted into an explicit UGTS proof problem and solved exactly where this release closes the full legal graph. No unsupported whole-game result is substituted for the remaining proof work.